

Panteon Strategy Game Demo

Project overview - architecture, design workflow and how to run it

Engine Unity 6.3 LTS (6000.3.24f1), URP 2D

Architecture FlowloC 1.22.4

Repository github.com/aliciinar/PanteonCase

Entry scene `Assets/Modules/MainModule/Scenes/MainScene.unity`

This document accompanies my submission for the Panteon Game Developer case. It explains how the game is built, how a designer can tune and test it without touching code, and why it is built on **FlowloC**. The last section goes into the details of how FlowloC is used in the project, for reviewers who want to follow the code.

Getting started

- Open the project with **Unity 6000.3.24f1** (Unity 6.3 LTS). Packages resolve on their own: FlowloC comes from the OpenUPM scoped registry listed in `Packages/manifest.json`.
- Open `MainScene` and press **Play** - or open **MainTestScene** to play on the designer's test configs (section 4). The entry scene path is in the header above.

Action	Input
Place a building	Click it in the production menu, drag the ghost on the board, confirm with the tick (or cancel with the cross)
Select a building or a soldier	Left click on it
Produce a soldier	Select a Barracks, click a soldier card in the information panel
Move / attack with the selected soldier	Right click on an empty cell / on a building or a soldier
Scroll the production menu	Mouse wheel or drag
Close a message	Click anywhere

1. Built on FlowloC

The whole project is written on **FlowloC**, a signal-driven inversion-of-control framework for Unity. A game is split into modules that each own their state, logic and presentation. Modules never reference one another: they talk through signals, and the only place two modules meet is a *Connector*. Every piece of work is a small *Command* bound into a readable sequence, state lives in *Models*, and scene objects are thin *Views* driven by *Mediators*.

I took part in the development of FlowloC. I know its internals, its tooling and the reasons behind its rules, which is why I chose it for this case: it let me spend the time on the game itself while keeping the code structured the way a production team would expect.

FlowloC on GitHub: github.com/FlowArc/FlowloC

Why FlowloC - the advantages in this project

Hard module boundaries

Every module has its own assemblies (Runtime, Shared, Signals). A module cannot even compile a reference to another module's internals - the compiler, not discipline, keeps the architecture clean.

Flows you can read in one place

Each context binds a signal to a sequence of commands. Reading the binding tells you exactly what an operation does, step by step, without opening every class.

Small, focused units

Decisions live in Commands, state in Models, scene references in Views. No MonoBehaviour god-classes, no singletons, no static state that survives between play sessions.

Loose coupling through Connectors

Features are wired declaratively. Replacing or removing a module means changing its connections - not hunting for call sites across the codebase.

Production services included

Screens (Addressables, layers, pooled instances), object pooling, loading sets and an update provider come with the framework, so none of them had to be written from scratch.

Data-driven by design

ScriptableObjects are filed on a Root's adapter and read through a shared data model, so configuration is swappable per scene - which is what makes the design test scene possible.

Tooling

Module generator, Module Scanner (checks folders, assemblies, references and namespaces), generated module cards (MODULE.md) and the Flow Console, which traces every signal and command.

Testable in isolation

Every module can ship a test module with its own scene, so a screen or a system can be run and checked on its own.

2. The game

A 2D strategy sandbox on a 10×10 grid of 32×32 px cells. The screen has three parts: the **production menu** on the left, the **game board** in the middle and the **information panel** on the right. Camera and UI adapt to the window's size and aspect ratio; the board is always fitted between the two panels.



The board with a Barracks, a Power Plant and the three soldier types (Kenney Medieval RTS art, CC0).

Features

- **Infinite production menu** - an endlessly scrolling list of buildings built on object pooling: only the cards on screen exist, and they are recycled as rows scroll in and out.
- **Placement** - picking a building shows a ghost on the nearest free area, coloured green where it fits and red where it does not, with its name and size (*Barracks 4×4*). It can be dragged and is confirmed or cancelled with the buttons above it.
- **Buildings** - Barracks (4×4, 100 HP) and Power Plant (2×3, 50 HP). New building types are added purely in data.
- **Production** - selecting a Barracks shows its image, health and the soldiers it produces. Clicking a card spawns that soldier, which walks out of the building's door to its spawn point - or, when that is taken or walled off, to the nearest free cell it can walk to (one **BFS** out of the door finds both the cell and the walk).
- **Soldiers** - three types, 10 HP each, dealing 10 / 5 / 2 damage. Left click selects, right click on an empty cell moves along the shortest path (**A***, routing around buildings).
- **Combat** - right click on a unit or building attacks it: one **A*** whose goal is any free cell next to the target walks the soldier to the side it can reach soonest, and it strikes. Hits flash the target and show a health bar; at 0 HP it is destroyed with a pooled explosion effect.
- **One action at a time** - while a unit is acting, the game is locked through a single runtime asset, so orders never overlap.
- **Feedback** - a selected building or soldier is highlighted on the board and shown in the information panel; an order that cannot be carried out (no way to the target or to any side of it, no free cell a new soldier can walk to) opens a message popup.



A selected Barracks: its health and the soldiers it produces.



A destroyed Power Plant explodes in pooled puffs.

Performance

- Everything that comes and goes is pooled: production cards, buildings, soldiers, unit cards and explosion effects.
- The board's cells are one Tilemap - a single mesh built once, not an object per cell - so a bigger grid costs nothing per frame.
- All game sprites are packed into two sprite atlases (board and UI), share one material and are dynamically batched. Measured in play: **10 batches / 9 SetPass calls** with the full board on screen (target: under 20).
- Nothing polls in `Update`: input is read only while a button is held, through FlowIoC's update provider, and window resizes are events rather than checks.

3. Designed for designers - ScriptableObject configuration

Every value a designer might want to change lives in a **ScriptableObject config** (`CD_` assets), never in code or scattered over prefabs. Changing the game's balance, look or feel is an edit in the Inspector.

Config	What it controls
<code>CD_Buildings</code>	Per building: name, menu icon, board sprite, footprint, health, the units it produces, its door and spawn point (edited on a visual cell map in the Inspector). For all buildings: selection colour, hit flash, health bar layout, placement ghost colours and the explosion (sprites, number of puffs, size, timing).
<code>CD_Units</code>	Per soldier: name, sprite, health, damage, walking speed, strike duration and lunge reach. For all soldiers: selection colour, hit flash, explosion, and the messages shown when an order is refused.
<code>CD_GameBoard</code>	Grid size, cell size in pixels, pixels per unit, frame padding and the cell sprite.
<code>CD_ProductionMenu</code>	Columns, card size and spacing of the production menu.
<code>CD_PopupScreen</code>	The popup's icon and its opening animation.

Runtime-only state is kept apart from configuration in `RD_` assets - for example `RD_Grid` shows the live board in the Inspector while playing, and `RD_GameStatus` holds the one-action-at-a-time lock.

4. The design test scene - MainTestScene

The game can also be launched from a dedicated test scene:

`Assets/Modules/MainModule/zTestModules/MainTestModule/Scenes/MainTestScene.unity` . It is the full game, but it runs on **test copies of every design config**, so a designer can experiment freely without changing what ships.

- **Test configs** - `CD_Buildings_Test` , `CD_Units_Test` , `CD_GameBoard_Test` , `CD_ProductionMenu_Test` and `CD_PopupScreen_Test` in `MainTestModule/Scriptables` . They are assigned to the scene's Roots as overrides; the shipped configs and `MainScene` are never touched.
- **A prepared start board** - `RD_Grid_Test` lists which buildings and soldiers stand on which cells when the game starts, everyone at full health. A designer can set up a situation once (a besieged soldier, a crowded board, a specific layout) and get it back on every Play.
- **How to use it** - open `MainTestScene` , edit the `_Test` assets or `RD_Grid_Test` , press Play. A separate test context checks that the scene really runs on the test copies and places the start board as soon as the board is built.



A refused order: the soldier cannot reach the surrounded target, and the popup says why.

5. Documentation in the project

Besides this PDF, every module carries its own card, `MODULE.md`, next to its code: what the module is for, the decisions whose reasons are not visible in the code, and its known gaps. Its lower half - assemblies, root, context, incoming and outgoing signals - is generated by FlowloC's Module Scanner, so it always matches the code. The same content is in the repository's `README.md`. The case brief itself is in `Assets/Resources`.

6. Known limitations

- Soldiers walk through one another on the way; each still stops on a cell of its own.
- The click that closes a message popup counts as a press on UI, so it also clears the selected soldier.
- Pool keys are strings, as FlowloC's pool expects; a typo in a key only shows at run time.
- The `_Test` configs are copies: a field added to a config later appears in its test copy with the default value.

7. What I would add next

- **Local avoidance** so soldiers keep apart while walking, and **box selection** to order several soldiers at once.
- **An opposing side** with a simple AI, so combat has a purpose beyond the sandbox.
- **Sound** for placement, strikes and explosions, and a **UI theme config** so panel and text colours are data too.
- **Pool keys as enums** per module, so a wrong key is a compile error rather than a run-time one.
- **Automated tests** for the grid's searches (BFS and A*) in the Grid module's test module.

8. Details - how FlowloC is used in this project

Module map

Module	Kind	Responsibility
MainModule	Core	Boots the game (loading screen, screen preloading, pool warm-up) and announces <code>Started</code> ; reports window resizes.
ScreenModule	Core	Hosts the screen manager; owns the generic PopupScreen .
ConnectorModule	Core	The only place modules meet: one sub-context per counterpart wires Outgoing signals to Incoming ones.
GridModule	System	Owns the grid and everything standing on it, health included; resolves what a click landed on; BFS / A* searches behind <code>IGridService</code> .
GameBoardModule	System	Draws the framed board as one Tilemap and announces its bounds.
BuildingsModule	System	Placement, board buildings, selection, destruction; hosts the ProductionMenuScreen .
UnitsModule	System	Spawning, selection, movement, combat and destruction of soldiers.
InputModule	System	The only reader of raw input; announces presses, drags and right clicks in world units.
CameraModule	System	Fits the board into the space left between the two panels.
GameplayModule	System	Owns the action lock; hosts the InformationScreen and InfoScreen .
LoadingModule	Service	Loading sets and progress for the boot, with its loading screens.

Signals and connectors

A module's public surface is its signal holder: `Incoming` is what it accepts, `Outgoing` what it announces. Connectors join them, and translate where the payload differs:

```
// UnitsConnectorSubContext - what the units module hears and what it tells others
_gridSignals.Outgoing.UnitPressed.Connect(_unitsSignals.Incoming.SelectUnit);
_gridSignals.Outgoing.OccupantSecondaryPressed.Connect(_unitsSignals.Incoming.AttackWithSelectedUnit);
_unitsSignals.Outgoing.UnitSelected.Connect(_infoScreenSignals.Incoming.ShowUnitInfo);
_unitsSignals.Outgoing.OrderRefused.Connect(_popupScreenSignals.Incoming.ShowPopup);
```

Flows are read from one context

An operation is a sequence of commands bound to a signal. Each command does one unit of work and hands its result to the next step; a command that can fail stops the sequence.

```
// UnitsSystemContext - an attack order, top to bottom
CommandBinder.Bind(_signals.Incoming.AttackWithSelectedUnit)
    .ToSequence<PlanAttackCommand>() // one A* to any free cell next to the target, or refuse
    .ToSequence<StrikeWithBoardUnitCommand>() // walk, lunge, strike (DOTween sequence)
    .ToSequence<SignalDispatchCommand>(_signals.Outgoing.ActionStarted); // locks the game
```

The whole attack, across modules: **Input** reports the right click → **Grid** finds what is under it and announces `OccupantSecondaryPressed` → **Units** plans and plays the strike → `AttackLanded` → **Grid** lowers the target's health and, at zero, removes it and announces `BuildingDestroyed` → **Buildings** returns the object to the pool with an explosion → the unit's action ends and **Gameplay** unlocks the game.

Rules the project follows

- **The grid owns the board.** A building's or soldier's data exists once, on the cells it covers; no other module keeps a copy. A cell leads to the data, and the data to its board object.
- **Models only for state that spans events** - the selected unit, the pending placement, the action lock. Everything else travels with the signal or from one command to the next.
- **Decisions belong in commands.** Contexts only declare bindings; views and mediators hold no game rules.
- **Configuration is filed, not dragged around.** A shared config is filed once on a Root's adapter and read through `ISharedDataModel` ; test Roots can file their own copies - the basis of `MainTestScene`.
- **Screens are pooled and addressable.** They are opened through `IScreenService` and filled by the command that opened them; a screen's prefab references no atlas sprite, so no bundle duplicates an atlas.
- **Services cross module boundaries directly.** `IGridService` , `IPoolService` and `IScreenService` are injected by interface; everything else crosses through a Connector.

Tooling used

- **Create Module** - every module, screen and test module was generated, so folders, assemblies, namespaces and log channels follow one shape.
- **Module Scanner** - checks every module's structure and references; the project reports **0 issues**.
- **Flow Console** - traces each signal and command as it runs, which makes a flow easy to follow and debug.

Art: Kenney Medieval RTS, Game Icons and Smoke Particles packs (CC0). Source code: github.com/aliciinar/PanteonCase · FlowloC: github.com/FlowArc/FlowloC